



Isothermal piston gas compression for compressed air energy storage

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ABSTRACT

Currently, Compressed Air Energy Storage systems mainly use adiabatic compression. Compared with isothermal compression, approximately twice the electricity is transformed into heat. Twice the heat passed into heat exchange mediums leads to twice the heat transfer losses. Enhancement of the heat transfer between air and environment to achieve isothermal compression is an effective approach to improve the turnaround efficiency of CAES systems. An isothermal piston structure is proposed to do isothermal compression. One end of the isothermal piston is connected to a traditional piston, and the other end dips into a liquid medium in the bottom of a cylinder. This forms a gas–solid–liquid three-layer heat transfer structure. A porous medium is used to enhance the heat transfer from the air to the liquid. As the heat capacity of the liquid is much greater than that of the air, the temperature of the liquid remains unchanged as well as the compressed air. A new method is proposed to look at the thermodynamics of the compressor with two dimensionless parameters Ka and Xu . Ka describes the extent of the compressor approaching isothermal. When Ka is over 80, the temperature of the air is reduced by 80% compared with adiabatic condition. In the case of using an aluminum porous medium, the compression efficiency increases by 11% at the compression ratio of 7 and the speed of 1200 r/min.

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1. Introduction

Integrating renewable energy sources (wind and solar energy) into electrical grid is one of the main challenges for renewable energy systems. It results from intermittent and unstable of renewable energy sources. The capacity of wind energy in China was more than 169 GW in 2016. The average utilization time of a wind plant is less than 1742 h/year. The abandoned wind energy reaches over 20% with an economic loss of over 10 billion RMB per year [1]. Fig. 1 shows the ratios of abandoned wind energy for wind plant in China [1,2].

Developing appropriate energy storage technologies is essential for the application of renewable energy sources [3,4]. Such technologies allow the storage of the unsteady energy from a variety of renewable sources to be recovered when needed in order to balance energy excesses or deficits and to improve the controllability of power plants.

Table 1 presents four types of energy storage technologies including mechanical energy storage, electromagnetic energy storage, chemical energy storage and thermal energy storage. Compressed

air energy storage (CAES) [3,4] is a form of mechanical energy storage that has many advantages: this system is suitable for large-scale applications (100 MWh, battery < 10 MWh), it is environmentally friendly (no heavy metal pollution), and has long service life (40–50 years, battery < 20 years) [6]. However, the following two main issues lead to its poor economic performance in commercial applications. The payback period for an investment in a CAES system is 8–10 years in China [7].

- (1) Low energy density: large amount of high-pressure air containment is required. Generally, a natural cavern is used as HP air containment for its low cost, but natural caverns are not always founded where they are needed. The costs of man-made steel tank is expensive [8].
- (2) Low efficiency of the system (turnaround efficiency < 60% [9], battery > 90%): CAES systems store electricity at valley charges and generate electricity at peak charges. Profits made with the peak-valley differences of charges depend on the turnaround efficiency of CAES systems.

Liquid air energy storage (LAES) was proposed [10] to improve energy density by liquefying air. The energy stored in a LAES system is mostly coolth exergy. The pressure of a liquid air containment is close to atmosphere. The volume of the liquid air contain-

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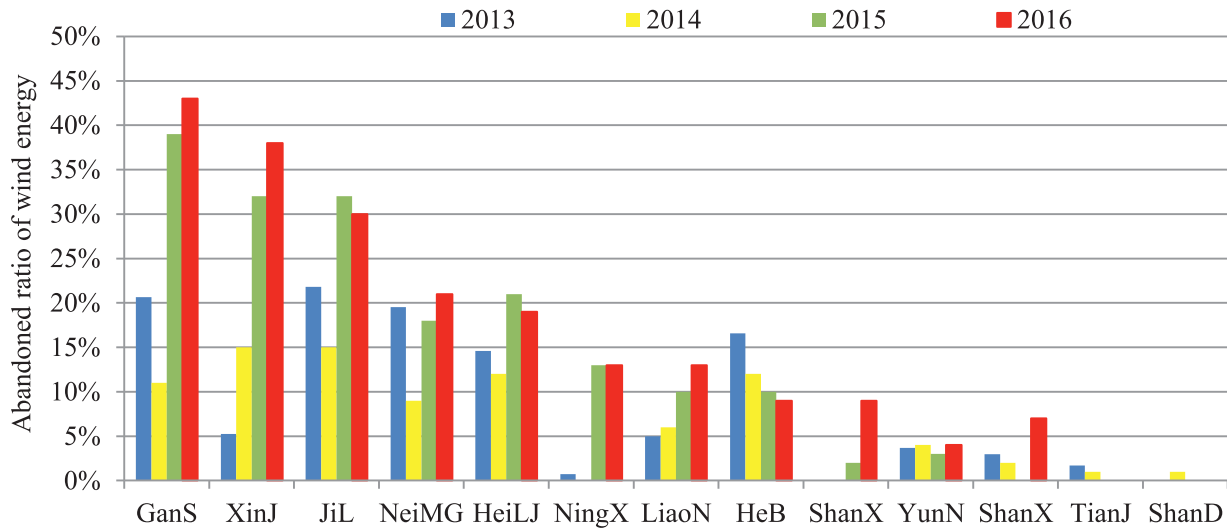


Fig. 1. The abandoned ratio of wind energy in 2013–2016.

Table 1

Characteristics of energy storage technologies [5].

Method		Efficiency (%)	Capacity (MWh)	Storage duration	Energy Density (Wh/L)
Electro-magnetic	Superconductor	95–97	0.001–0.015	Seconds–hours	0.2–2.5
	Supercapacitor	70–80	0.0005	Seconds–hours	10–30
Chemical	Lead acid	80–90	0.001–40	Minutes–days	50–80
	Lithium Ion	90–97	0.004–10	Minutes–days	200–500
Thermal	Pumped thermal	30–60	–	Minutes–days	80–500
Mechanical	Pumped hydro	70–80	500–8000	Hours–months	0.5–1.5
	CAES	42, 54 [8]	100–1000	Hours–months	3–6
	LAES	55–80	2.5	–	4–6 times than CAES

Nomenclature

p	Pressure of air (Pa)
Δp	Pressure loss of fluids (Pa)
V	Volume of air (m^3)
m	Mass (kg)
θ	Temperature (K)
U	Internal energy (J)
Q	Heat (J)
Q_{air}	Heat transferred to air (J)
Q_{liq}	Heat transferred to liquid (J)
W	Work (J)
W_{iso}	Work of isothermal compression (J)
W_{adi}	Work of adiabatic compression (J)
S	Surface area (m^2)
S_{air}	Air-Porous medium contact area (m^2)
S_v	Specific Surface Area (m^2/m^3 or cm^2/cm^3)
S_{vex}	Heat transfer specific surface area (m^2/m^3 or cm^2/cm^3)
R	Ideal gas constant (J/kg K^{-1})
C	Heat capacity (J/kg K^{-1})
C_v	Heat capacity of air (J/kg K^{-1})
κ	Heat capacity ratio (1.4)
h	Heat transfer coefficient ($\text{J/m}^2 \text{K}^{-1} \text{s}^{-1}$)
η	Efficiency (%)
L	Piston Stroke (m)
l	Length of a porous medium
t	Time (s)
T_p	Time for piston to compress (s)
u	Piston speed (m/s)

n	Rotational speed (r/s, r/min)
r_v	Volumetric compression ratio
ρ	Density (kg/m^3)
μ	Dynamic viscosity of fluids (Pa s)
r_p	Porosity of a porous medium
d	Hydraulic radius [m]

Subscripts

air	Air
por	Porous medium
liq	Liquid medium
0	Initial condition
a	Atmosphere

Superscript

$*$	Dimensionless variable
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ment of a LAES system is 10 times smaller than that of a CAES system. The cost could be inexpensive for commercial use. The world's first LAES pilot plant (350 kW/2.5 MWh) was built in 2012 [11]. Challenge of this system will be managing the coolth exergy at -196°C and the heat generated from liquefaction to obtain an acceptable turnaround efficiency. There are two main methods for improvement: (1) using cascading cycles to reduce the temperature gradient of heat transfer and associated exergy loss [12,13], and (2) integrating LAES system with other power sources and energy storage system [14,15].

The compression of a CAES system [7] is close to adiabatic. The heat generated from air compression cannot be sufficiently transferred with its environment. This results in the rise in the temperature and the pressure of the air. Accordingly, more work performed in resisting the force of the air is converted into heat. Approxi-

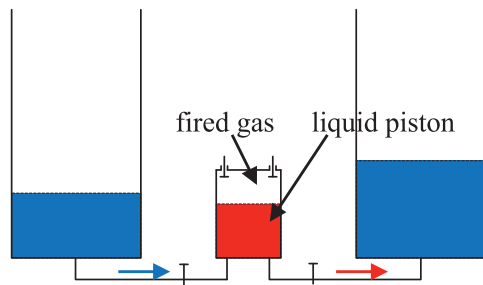


Fig. 2. liquid piston working principle.

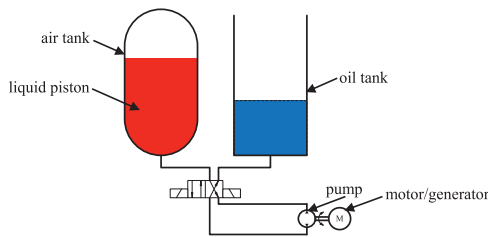


Fig. 3. Closed-loop liquid piston system.

mately half of the electricity is converted into heat and exhausted [16].

There are two main approaches to improve the turnaround efficiency of CAES system: (1) using the adiabatic compression and recycling the heat generated from air compression, and (2) using the isothermal compression to reduce the heat generated from air compression (Fig. 5).

Isothermal compressed air energy storage (ICAES) has two research directions. The first one is to use water sprays to cool compressed air. Coney [17] injected water into a compressor to cool the compressed air. The volume of the compression chamber was 46 liters and the compression ratio was 25. The experimental results showed that the temperature of working air was below 100 °C (adiabatic 500 °C), and compression work reduced by 28%. Qin [18] injected 30 μm diameter water mist into a three-stage compressor. The simulation results revealed that the compression efficiency could reach 89% (adiabatic 26.7%). LightSail [19] injected water spray into air storage tanks, in which the compression heat was absorbed and stored. The reported temperature was less than 10 degrees. The thermodynamic efficiency was 90% and the turnaround efficiency was 70%. SustainX [20] produced water foam and injected into outside compression/expansion cylinders through a port. Compared with water spray, the foam requires less energy for generating and longer effective time for cooling.

The reported spray cooled isothermal compression used a pump to generate 3–5 MPa pressurized water. The pressurized water speeded up in a nozzle and was atomized when injected into the surrounding air. The energy consumption of the pump accounted for 10% of total compression [21].

The second method is to use a liquid piston. Liquid pistons were used in internal combustion engines for pumping water known as the Humphrey pump (Fig. 2) [22] that was dated back to 1906. The liquid piston can be applied to an air compressor (Fig. 3). The pump is driven by the motor to force the liquid into the compression chamber. As the liquid level in the compression chamber rises, the volume decreases and the pressure increases. The injected liquid had the same functions as a conventional solid piston, therefore it is referred to as a liquid piston. As liquid is flexible and is able to adapt to the irregular shape of the compression chamber, the geometry of the compression chamber could be designed to increase the heat transfer area.

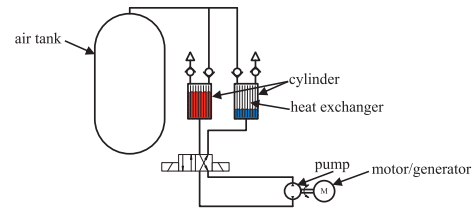


Fig. 4. Open-cycle system.

In 2002, Reller [23] proposed a closed-cycle hydraulic-pneumatic energy storage system (C-HyPES). In Fig. 3, the pump is used to inject liquid (hydraulic oil) into the air tank. The air stored in the air tank is compressed by the liquid piston. When generating electricity, the gas expands to drive pump turbine (P/T) in reverse with the liquid piston. The main disadvantage of C-HyPES is low energy density. The liquid tank had the same volume as the air tank, which largely increase the system's volume and weight. C-HyPES is still at the laboratory research stage and cannot be commercially used.

In order to improve energy density, Cyphelly [24] proposed a closed-cycle hydraulic-pneumatic energy storage system (O-HyPES), which combine the advantages of high power density of traditional compressors and high heat transfer rate of the liquid piston. In the closed system, the air is compressed in the air tank. The liquid should fill the air tank in order to complete the compression. As a result, the liquid's volume should equal to the air tank as well as the oil tank. In the open system (Fig. 4), the air is compressed not in the air tank, but in a much smaller cylinders. The liquid is circulated between the two cylinders and do not have to fill the air tank. Thus the large oil tank is not required and system's size does not increase with the air's volume. Therefore, the system's size is largely reduced. The first experimental storage plant of 2 MW capacity has been in trial operation in Texas, USA since 2012 [25].

Zhang [26] proposed an interrupted-plate heat exchanger to improve the heat transfer of an ICAES system. The heat exchanger is made of layers of stacked plates. The adjacent layers are perpendicular to one another. The influence of geometry on the heat transfer was analyzed. The reported results show decreasing the plate length or plate distance helps improve the efficiency.

Zhang [27] proposed a one-dimensional calculation model to optimize the axial porosity distribution of porous inserts of an ICAES system. Taking the compression/expansion chamber pressure as the objective function, the distribution of the inserts' porosity is obtained.

The compression efficiency could be reduced if the piston moves at a uniform speed, as the heat is not generated at a uniform speed. In one compression cycle, as the pressure continues to increase, more heat is generated at each end of a cycle compared with this is in the beginning of a cycle, i.e., the heat is not generated at a uniform rate. Thus, the piston's speed should be designed to maximize the heat transfer and minimize the friction. Saadat [28] used a Pareto method to optimize the piston trajectory by adjusting the valve opening that is related to the speed of the liquid piston.

James and Li [29,30] compared heat transfer and friction of a liquid piston and a conventional reciprocating piston by experiments. The reported energy loss of the liquid piston reduced by 19%. Considering friction loss, the total efficiency of the liquid piston reached 83%, which is higher than that of the reciprocating piston (70%).

Wieberdink [31] designed an ICAES system with a two or three stage compressor to validate an interrupted plate porous medium is effective at a pressure up to 210 bar. The reported result shows

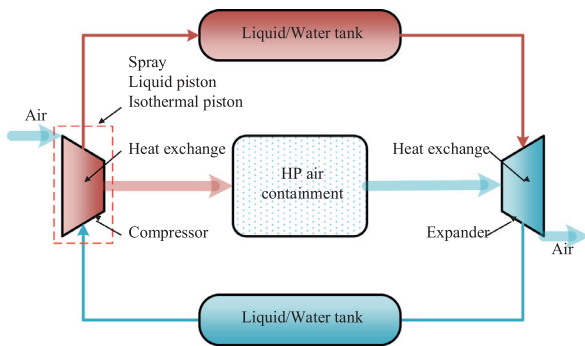


Fig. 5. Schematic of an ICAES system.

it can increase power density by 10 times without decreases in efficiency.

Flexibility of the liquid piston allows it to be adapt to irregular shapes of the compression chamber and the heat transfer mediums. This allows heat transfer structures to be complexly designed. The heat generated from air compression is sufficiently transferred from the air to the environment with liquid mediums. This reduces the temperature of the air and the compression work. However, there are still several issues with the liquid piston system. Firstly, as the air is in direct contact with the liquid, the air is unavoidably mixed with the liquid and circulates in the hydraulic system. This reduces modulus of elasticity of the liquid, increases the noise and the vibration of the hydraulic pump and pipeline, and results in cavitation in low pressure areas of the system. Secondly, adding a hydraulic system to the air compression system results in an increase in the overall system size and weight.

2. Isothermal piston and isothermal compressed air storage

2.1. Isothermal compressed air storage

Like general CAES systems, an ICAES system (Fig. 5) stores electricity at the off-peak period of electrical grids, and give it back at the peak period of electrical grids. During the off-peak period, air at ambient pressure is compressed in a compressor. The high-pressure air from the compressor is stored in a HP air containment. The off-peak electricity from grids is converted to the energy in the form of high-pressure air. During the peak period, HP air is released from the HP containment to produce electricity with an expander. Main differences between isothermal CAES and adiabatic CAES is the heat transfer of the air. In cases of adiabatic CAES, the temperature of the air increases significantly during air compression. High-temperature air exchanges heat with thermal stores outside compressors. In cases of ICAES, the heat transfer is taken place inside compressors. This allows the temperature of the air to be close to ambient during air compression. The heat generated from the isothermal compression is reduced. The energy loss related to the heat transfer in thermal stores decreases. All of the references (spray/liquid piston) discussed methods to improve heat transfer inside of the compressor with liquid fluids. The liquid fluids flows between the compressor and the tank to transmit heat between them.

2.2. Concept of the isothermal piston

The isothermal piston applies porous medium and liquid medium to form a gas-solid-liquid three-layer structure to enhance heat transfer. The porous media has an enormous surface area that increase the air-solid heat transfer area. The liquid medium is flexible to flow through inner passages of the porous media to en-

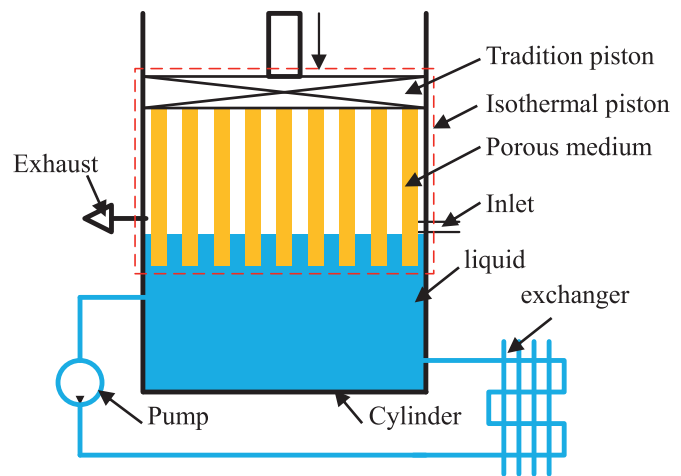


Fig. 6. Schematic of an isothermal piston.

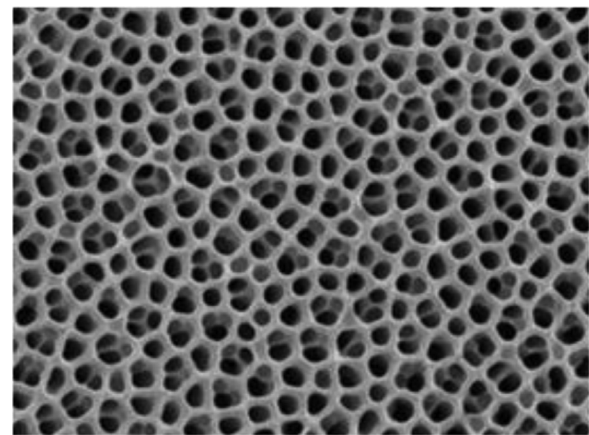


Fig. 7. Porous medium.

hance solid-liquid heat transfer. The compression heat is transferred through the porous media to the liquid medium. Finally, the heat held by the liquid medium is dissipated to environment by an external heat exchanger. As heat capacity of the liquid medium is much larger than that of the compressed air, the liquid's temperature almost remains unchanged even after it absorbs the compression heat.

As shown in Fig. 6, the isothermal piston compression system consists of a piston, a porous medium (Fig. 7), a cylinder, a pump, and an external exchanger. Both the air and the liquid are contained in the cylinder. The rod drives the piston to move up and down. The porous medium is integrated with the piston to form the proposed isothermal piston. It moves with the piston and is immersed into the liquid. As the addition of porous media greatly improves the heat transfer, the gas temperature is close to that of the liquid. The cylinder wall, the pump and the external exchanger form a heat dissipation circuit. As the liquid heat capacity is much larger than the gas, the increase in the liquid temperature is not much. The liquid circulates between the cylinder and the external exchanger to keep its temperature close to the atmosphere.

The working procedure is designed as follows:

I Intake stroke: the air inlet opens and the exhaust port is closed when the piston reaches the Bottom Dead center. The rod drives the piston to move up. The air from the atmosphere is forced a drawn into the cylinder until the piston reaches its Top Dead center.

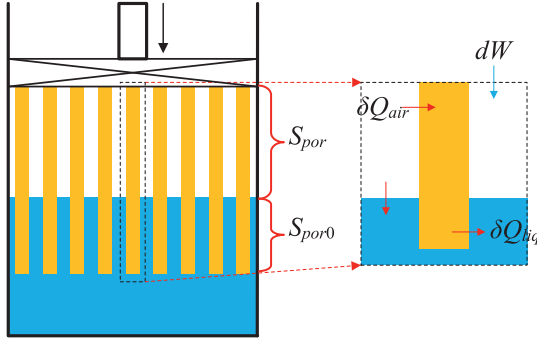


Fig. 8. Heat transfer in cylinder.

- II Compression stroke: the air inlet and the exhaust port are closed when the piston moves downwards. The air in the cylinder is compressed by the piston. Porous medium is immersed into the liquid. The compression heat is rapidly transferred to the porous medium.
- III Exhaust stroke: the exhaust port is closed when the piston reaches the Bottom Dead center. The air is discharged.
- IV The liquid circulation process: the pump is started when the liquid temperature rises to a setting value. The liquid is discharged from the cylinder and flows into the external exchanger for cooling. The air temperature drops close to atmosphere.

3. Calculation methods

3.1. Pressure

When the piston moves downward, the air pressure rises from the top to the bottom of the working chamber. The time it takes for the pressure wave to travel from the top to the bottom is much less than the compression time. Therefore, the pressure difference between the upper and lower sides can be neglected, i.e., the pressure is evenly distributed in the compression chamber.

When the air is compressed, the working chamber becomes small leading to rise of the air pressure. According to the ideal gas law, the pressure is related to the temperature and volume,

$$pV = mR\theta \quad (1)$$

3.2. Temperature

The work done by the piston (dW) is transformed to internal energy of the air (dU) and heat transferred to the surroundings (δQ). According to conservation of energy, the balance equation of the internal energy, the work and heat transfer is obtained,

$$dU = \delta Q + dW \quad (2)$$

According to the definition of specific heat (C), internal energy (dU) is expressed as product of the temperature ($d\theta$), mass (m) and specific heat (C):

$$dU = mCd\theta \quad (3)$$

According to Newton's law of cooling, the heat transfer (δQ) from object 1 to object 2 is proportional to contact area (S), and the objects' temperature difference ($\theta_1 - \theta_2$):

$$\delta Q = hS(\theta_1 - \theta_2)dt \quad (4)$$

where h is the heat transfer rate.

As shown in Fig. 8, applying the energy Eq. (2) to the air, the air's internal energy (dU_{air}) is obtained as following:

$$dU_{air} = -\delta Q_{air} + dW \quad (5)$$

Applying the energy Eq. (2) to the porous medium, the heat (δQ_{air}) absorbed from the air is transformed to the porous medium's internal energy (dU_{por}) and the heat transferred to the liquid (δQ_{liq}).

$$dU_{por} = \delta Q_{air} - \delta Q_{liq} \quad (6)$$

3.3. Work and efficiency

The external force is balanced with the air force of the air in the working chamber. The work done on the air is expressed as the volume integral of the pressure.

$$W = \int_{V_0}^V p dV \quad (7)$$

The work done to compress air is influenced by temperature of the working air. According to the ideal gas law, if the air temperature decreases, its pressure decreases as well. Accordingly, less force is needed to move the piston. Thus, if the temperature is not increased, i.e., isothermal compression, the compression work will reach its minimum. When applying Eq. (7) to isothermal compression, the work is expressed as follows:

$$W_{iso} = m_{air} R \theta_a \ln \frac{p}{p_a} \quad (8)$$

Similarly, if the air temperature is increased to its maximum, i.e., adiabatic compression, the compression work will reach its maximum. When applying Eq. (7) to adiabatic compression, the work is expressed as follows:

$$W_{adi} = m_{air} R \theta_a \left[\left(\frac{p}{p_0} \right)^{\frac{\kappa-1}{\kappa}} - 1 \right] / \left(\frac{\kappa-1}{\kappa} \right) \quad (9)$$

For isothermal compression, the compression work is completely converted to mechanical energy (exergy) that is stored in the air. For practical compression, the temperature will inevitably rise, and the compression work will be more than the minimum value (W_{iso}). The extra work is transferred into heat and dissipated during the cooling process. In this study, it is assumed that this part of the compression work is completely dissipated. Thus, the compression efficiency is defined by Eq. (12),

$$\eta = 1 - \frac{W - W_{iso}}{W} \quad (10)$$

3.4. Resistance

Although liquid-porous medium has advantages in improving the heat transfer, movements between the fluids (air and water) and the porous medium will result in fluid resistance, i.e., it introduces an additional resistance besides conventional piston. Its influence on the compression efficiency should be investigated. The fluid resistance consists of the viscous resistance and the inertia resistance [24]. The resistance-resulted pressure loss (Δp) by unit of length (l) of a porous medium is obtained as following:

$$\frac{\Delta p}{l} = \frac{\mu}{K} \cdot u + \rho \cdot C \cdot u^2 \quad (11)$$

where, μ is dynamic viscosity, ρ is density of fluids, u is relative velocity between a fluid and a porous medium, K is permeability of porous medium, and C is inertia coefficient of porous medium (or shape factor). K and C is defined as:

$$K = a \cdot \frac{r_p^3 d^2}{(1 - r_p^2)} \quad (12)$$

$$C = b \cdot \frac{(1 - r_p^2)}{r_p^3 d} \quad (13)$$

Table 2

Standard quantities and dimensionless variables.

	Pressure	Volume	Temperature	Heat Transfer Area	Time	Work
Standard quantities	p_a	V_0	θ_a	S_{por}	$T_p = L/u$	$p_a V_0$
Dimensionless variables	$p^* = p/p_a$	$V^* = V/V_0$	$\theta^* = \theta/\theta_a$	$S^* = S/S_{por}$	$t^* = t/T_p$	$W^* = W/(p_a V_0)$

where, a is a constant related to viscosity, b is a constant related to inertia, r_p is porosity, and d is the hydraulic radius of a porous medium.

Substituting Eqs. (12) and (13) into Eq. (11), the classical Ergun Equation [32] that models the resistance of the porous medium is obtained,

$$\frac{\Delta p}{l} = a \cdot \frac{(1 - r_p^2)\mu}{r_p^3 d^2} \cdot u + b \cdot \frac{(1 - r_p^2)}{r_p^3 d} \cdot \rho \cdot u^2 \quad (14)$$

Air resistance is negligible according to Eq. (14) as viscosity and density of air (17.9×10^{-6} Pa s, 1.18 kg/m^3) is much smaller than that of water (1.01×10^{-3} Pa s, 1000 kg/m^3). All the following discussions are about the liquid resistance. Air resistance is ignored.

Accordingly, work done by the liquid resistance and the efficiency is obtained respectively, as following:

$$W_{resistance} = \int_{V_0}^V \Delta p_{liq} \cdot dV \quad (15)$$

$$\eta_{+resistance} = 1 - \frac{W + W_{resistance} - W_{iso}}{W + W_{resistance}} \quad (16)$$

3.5. Dimensionless equations

Table 2 shows standard quantities used to obtain dimensionless equations. Applying Eqs. (1) and (3) to the air and dividing Eq. (5) by $m_{air} C_v \theta_a$ at both sides, the dimensionless temperature is obtained as following:

$$\frac{d\theta_{air}^*}{dt^*} = (\kappa - 1)p^* - Ka(\theta_{air}^* - \theta_{por}^*)S_{air}^* \quad (17)$$

where,

$$Ka = \frac{R}{C_v} \cdot \frac{h_{air} \cdot S_{por} \cdot \theta_a \cdot T_p}{p_a \cdot V_a} \quad (18)$$

Let the heat transfer rate be zero, Eq. (17) becomes the equation for the adiabatic compression. The first term in the right side shows the temperature differential of the air related to air compression. The second term expresses the porous medium's influence on the temperature differential. The term $(h_{air} \cdot S_{por} \cdot \theta_a \cdot T_p)$ in Ka measures the heat transfer between the air and the porous medium. The term $(p_a \cdot V_a)$ measures the work done by the piston or the heat generated from air compression. Ka [33] expresses the ratio between the heat transferred from the air and the heat generated in the air. When Ka is close to 0, the generated heat cannot be transferred to the porous medium effectively. The temperature of the air will increase like the adiabatic compression. When Ka is close to ∞ , all of the generated heat will be transferred to the porous medium. The temperature of the air will not increase like the isothermal compression. Numbers of Ka describe the level of the compression to be close to isothermal.

Dividing differentiate form of Eq. (1) by $p_a V_0$ at both sides, the dimensionless pressure is obtained as following:

$$\frac{dp^*}{dt^*} = \kappa \frac{p^*}{V^*} - Ka \frac{(\theta_{air}^* - \theta_{por}^*)S_{air}^*}{V^*} \quad (19)$$

The first term in the right side is pressure differential of the air in the adiabatic condition. The second term expresses the porous medium's influence on the pressure differential. An increase in Ka will result in less pressure rise.

Applying Eqs. (1) and (3) to the porous medium and dividing Eq. (5) by $m_{por} C_{por} \theta_a$ at both sides, the dimensionless temperature is obtained as following:

$$\frac{d\theta_{por}^*}{dt^*} = Xu[S_{air}^*(\theta_{por}^* - \theta_{air}^*) - (S_{por0}^* + 1 - S_{air}^*)(\theta_{por}^* - 1)] \quad (20)$$

where,

$$Xu = \frac{h_{liq} \cdot S_{por} \cdot \theta_a \cdot T_p}{m_{por} \cdot C_{por} \cdot \theta_a} \quad (21)$$

The term $(h_{liq} \cdot S_{por} \cdot \theta_a \cdot T_p)$ in Xu measures the magnitude of the heat passed through the porous medium. A greater change of the temperature of the porous medium will be made by passing more heat through the porous medium. $(m_{por} \cdot C_{por} \cdot \theta_a)$ expresses the magnitude of the heat capacity of the porous medium. A greater amount of the heat capacity will make the temperature of the porous medium more stabler. These two terms play an opposite role in changing the temperature of the porous medium. Xu is proposed to measure the heat passed through the porous medium per unit of the heat capacity by transferring these two terms into one ratio (heat pass-through vs heat capacity). When numbers of Xu increase, the change of the temperature of the porous medium will become greater.

3.6. Code validation

All the following reported results are obtained by numerically solving the differential Eqs. (17), (19) and (20). The ode45 solver in Matlab is used in this study. A fourth order Runge-Kutta method provides solutions. A fifth order Runge-Kutta method is for error control. The maximum step size is set to 0.005 s. For each simulation, it takes about 100 steps. Numerical solutions are compared with the analytical solution for adiabatic compression to validate the code. Fig. 9 shows that accuracy of this ode45 solver is acceptable for this study.

4. Results

4.1. Porous medium's equilibrium temperature

The heat accumulated in the porous medium will cause its temperature to rise, after it absorbs the compression heat. When its temperature is higher than the liquid's temperature, portion of the heat will be transferred from the porous medium to the liquid. After the compressor works for many cycles, the air-porous heat transfer will be balanced with the porous-liquid heat transfer. The temperature of the porous medium will become stable. This temperature is defined as equilibrium temperature. When the equilibrium temperature increases, the air-porous temperature difference becomes smaller and so as the heat transfer.

According to Eq. (20), the air-porous heat transfer area (S_{por0}^*) affects the equilibrium temperature. Parameter S_{por0}^* is the ratio of the porous medium's surface area below the liquid surface to that above the liquid surface, when the piston is at the Top Dead Center. As shown in Fig. 10, the equilibrium temperature is reduced as S_{por0}^* increases. The equilibrium temperature declines below 1.01, when S_{por0}^* is 0.4 ($Ka = 100$). The equilibrium temperature grows with an decrease in Ka . When Ka declines to 1, to keep the equilibrium temperature below 1.01 the value of S_{por0}^* should increase to 8.

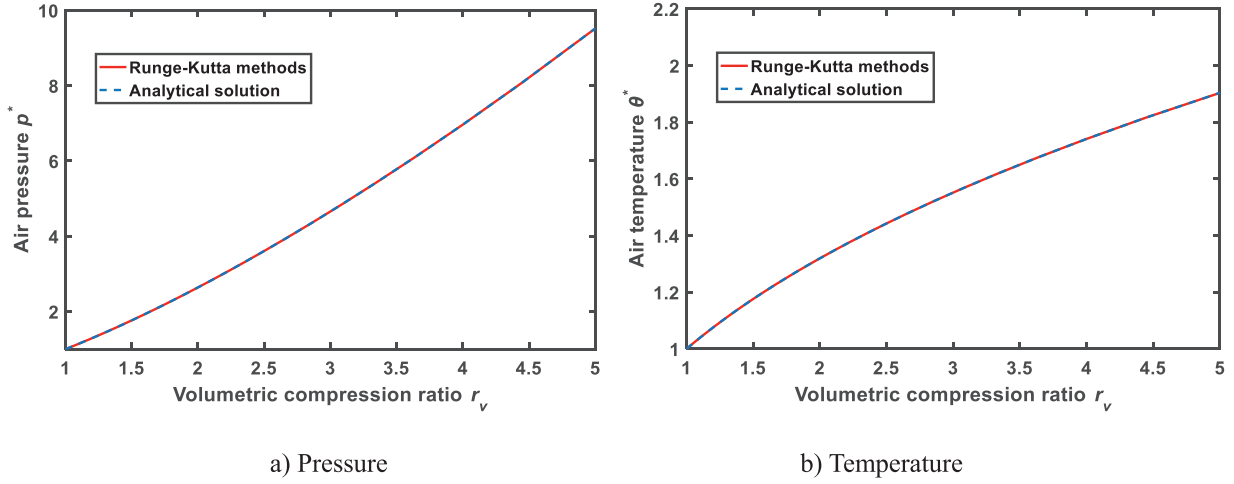
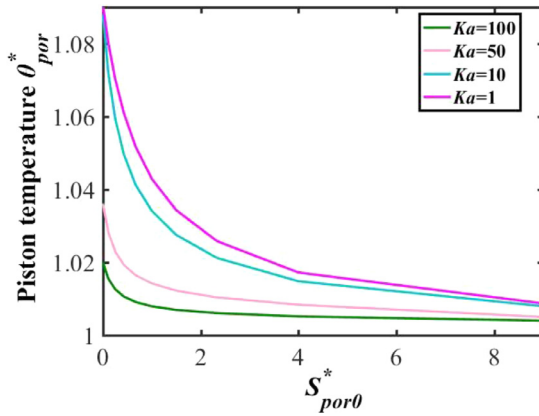
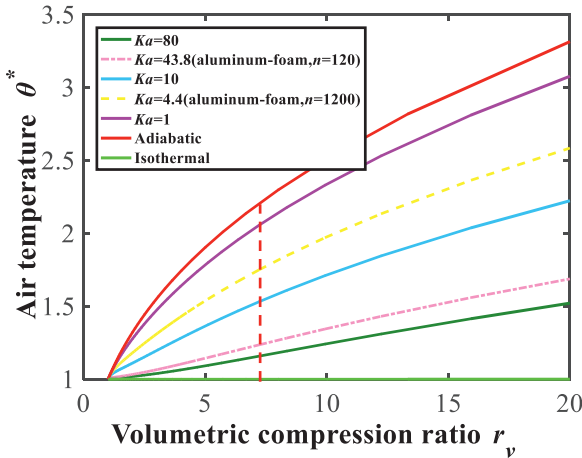


Fig. 9. Code validation.

Fig. 10. Influence of S_{por}^* on the porous medium's temperature.Fig. 11. Influence of parameter Ka on the temperature of the air.

4.2. Dimensionless parameter Ka

Numbers of Ka mainly affects the temperature of the air. Fig. 11 shows the curves of air temperature at different compression ratio. The temperature curves lie between the adiabatic curve (red) and isothermal curve (green). As values of parameter Ka increase from 1 to 80, the curves gradually approach the isothermal curve. Form of Ka in Eq. (18) is changed with parameters of the porous

medium so that influence of properties of the porous medium can be studied in the following discussion.

$$Ka = \frac{h_{air}}{\rho_a C_v} \frac{S_v}{r_p} \frac{1}{2n} \quad (22)$$

where, S_v and r_p are respectively the Specific Surface Area and the porosity of porous mediums. n denotes rotational speed of the compressor. The cycle period of the compressor equals $1/n$. Time (T_p) for compressing is approximately half of the cycle period $1/(2n)$. Volume of the cylinder (V_{cy}) comprises volume of the porous material and volume of the air V_a (void space). Specific Surface Area (S_v) is defined as surface area (S_{por}) per unit volume (bulk volume V_{cy}). r_p is defined as fraction of the volume of voids (V_a) over the total volume (V_{cy}). (S_{por}/V_{cy}) in Eq. (18) is transformed to (S_v/r_p). [r/s] is the unit of rotational speed n in Eq. (22) because "s" is the SI unit for time. [r/min] is used in the following discussion, because it is often used for compressors.

Applying the aluminum material [34] (parameters described in Table 2) to Eq. (22), the value of Ka is 4.4. When the aluminum material is used, the result shows the air temperature (pink curve, $r_v = 7$, $n = 1200$) is reduced by 18% compared with the adiabatic condition. When Ka is over 80, the air temperature could be reduced by 80%.

Table 3,4

According to Eq. (19), Ka not only affects the temperature but also the pressure. Fig. 12 shows the curves of air pressure at different compression ratio. As Ka increases, the air pressure decreases and approaches the isothermal curve. When the aluminum material is used, the result shows the air pressure (pink curve, $r_v = 7$, $n = 1200$) is reduced by 16% compared with the adiabatic condition.

Fig. 13 shows the variation of compression efficiency with compression ratio. The y-axis represents the growth of the compression efficiency of the isothermal piston compared with the adiabatic condition. The red curve and the green respectively represent the results of adiabatic compression and isothermal compression. Other curves represent the results of isothermal piston compression. As the compression ratio increases, the growth of the efficiency becomes greater. When the compression ratio is between 1 and 7, the compression efficiency increases logarithmically with the compression ratio. When the compression ratio is greater than 7, the increase in compression efficiency tends to be saturated. As Ka increases, the compression efficiency gradually increases and approaches to isothermal compression. When the aluminum material is used, the result shows the compression efficiency (pink

Table 3
Porous material parameters.

Porous material	Specific heat/ C_{por}	density/ ρ_{por}	Heat transfer specific surface area/ S_{vex}
Aluminum [34]	880 J/(kg °C)	2700 kg/m ³	8.6 cm ⁻¹ 18.7 cm ⁻¹ 29.78 cm ⁻¹

Table 4
Simulation parameters.

Section	S_{por0}^*	Ka	Xu	h_{air} w/(m ² •K)	S_{vex} cm ⁻¹	n r/min
4.1	0–9	1–100	0.01	–	–	–
4.2	8	1–80	0.01	–	–	–
4.3	8	50	0.01–10	–	–	–
4.4	8	–	–	30* [21]	8.6–29.8	120
4.5	8	–	–	30* [21]	29.8	100–2200

* The heat transfer rate is tested in a similar condition and the reported value ranged from 10 to 120 w/(m² K).

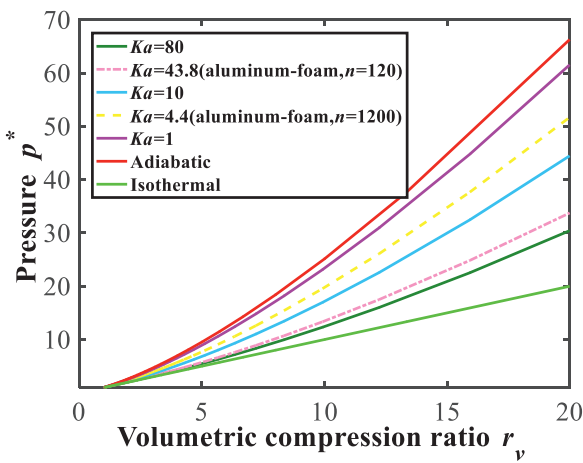


Fig. 12. Influence of parameter Ka on the air pressure.

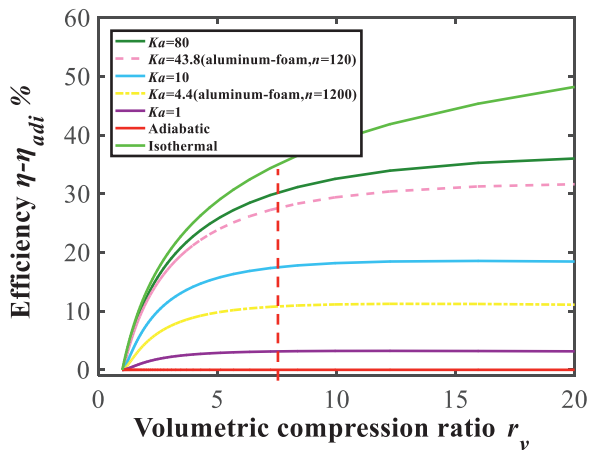


Fig. 13. Influence of parameter Ka on compression efficiency.

curve, $r_v = 7$, $n = 1200$) increases by 11% compared with the adiabatic condition (86.1%).

4.3. Dimensionless parameter Xu

Parameter Xu affects the temperature variation of porous medium. As shown in Fig. 14, the temperature of the porous medium waves around the equilibrium temperature. When parameter Xu varies in the range of 0.1–10, the temperature and pres-

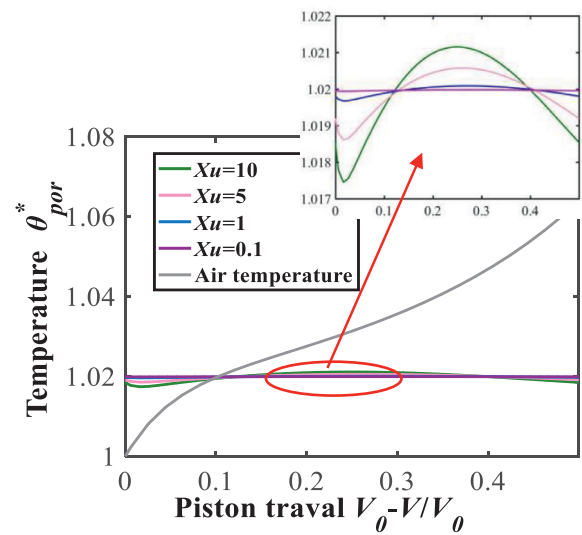


Fig. 14. Influence of parameter Xu on the porous medium's temperature.

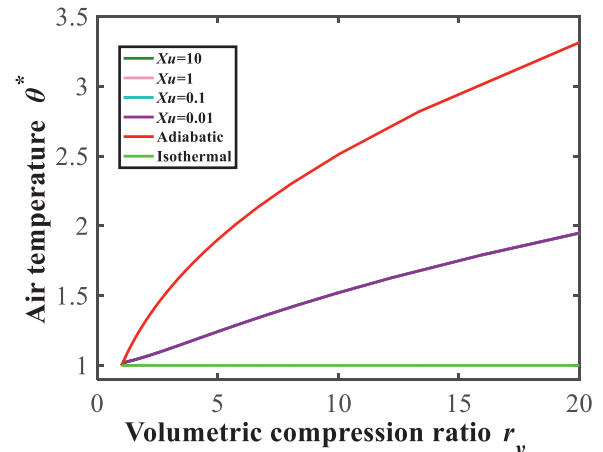
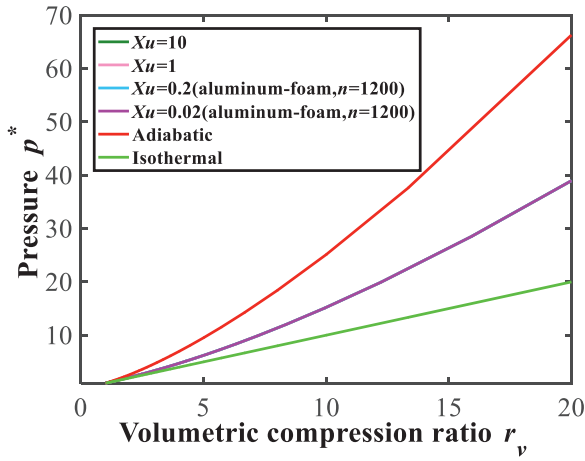
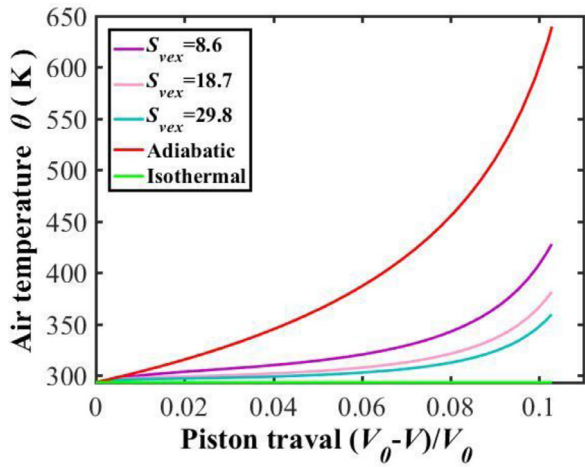


Fig. 15. Influence of parameter Xu on the air temperature.

sure curves in Figs. 15 and 16 are almost overlapped, i.e., parameter Xu has little effect on air pressure and temperature. According to Eq. (20), Xu directly determines the temperature of the porous medium (θ_{por}^*). And parameter Xu indirectly influences the air temperature and pressure by the variation of θ_{por}^* , according to Eqs. (18) and (19). As shown in Figs. 14 and 15, the variation range

Fig. 16. Influence of parameter Xu on the pressure.Fig. 17. Influence of heat transfer surface area S_{vex} on the air temperature.

of the porous medium's temperature θ_{por}^* is an order smaller than that of air θ_{air}^* , i.e., the temperature difference ($\theta_{air}^* - \theta_{por}^*$) is approximately equal to θ_{air}^* . Therefore, according to Eqs. (18) and (19), the influence of parameter Xu on air pressure and temperature could be neglected. Applying the aluminum material to Eq. (23), values of parameter Xu are 0.02 and 0.2 respectively.

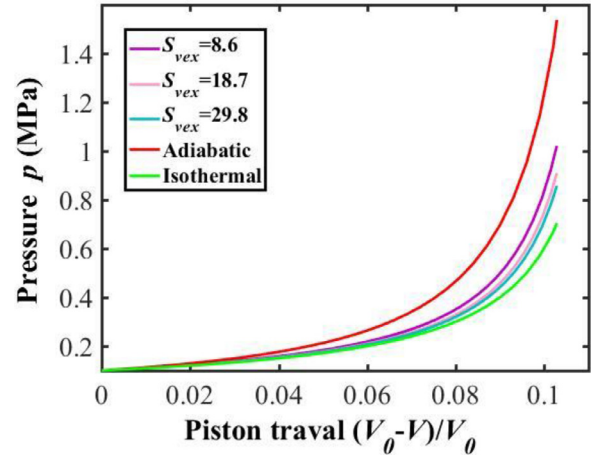
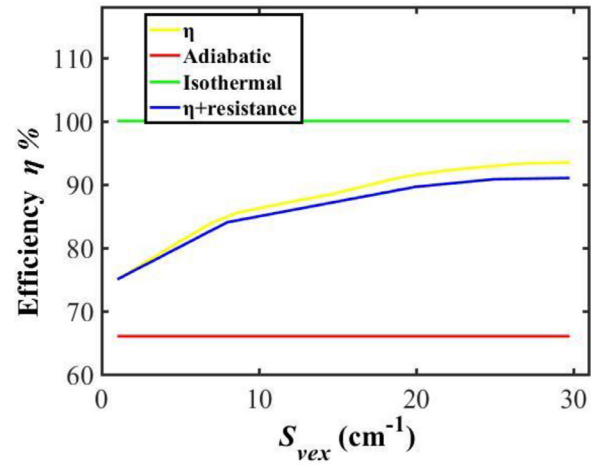
$$Xu = \frac{h_{liq}}{\rho_{por} C_{por}} \frac{S_v}{1 - r_p} \frac{1}{2n} \quad (23)$$

4.4. Heat transfer specific surface area S_{vex}

According to Eq. (22), parameter Ka is mainly determined by rotating speed, specific surface area and porosity. The specific surface area and the porosity are not a set of independent quantities, i.e., change of the porosity will cause the specific surface area to change. The heat transfer specific surface area is defined as the ratio of the specific surface area to the porosity to investigate the influence of the porous medium on compression.

$$S_{vex} = \frac{S_v}{r_p} \quad (24)$$

Applying the aluminum material [25] to Eq. (24), the heat transfer surface area is 8.6 cm^{-1} , 18.7 cm^{-1} and 29.8 cm^{-1} . Fig. 17 shows the temperature curves. The red curve and green curve represent the curves in adiabatic and isothermal condition respectively. When the compression ratio is 7, the temperatures of the compressed air are 639.7 K and 293 K. When the porous medium

Fig. 18. Influence of heat transfer surface area S_{vex} on the air pressure.Fig. 19. Influence of heat transfer surface area S_{vex} on the compression efficiency.

is used, the temperature are reduced to 427.7 K, 381.1 K and 359.3 K respectively.

Fig. 18 shows the pressure curves. The red curve and green curve represent the curves in adiabatic and isothermal condition respectively. When the compression ratio is 7, the pressure of compressed air is 1.5 MPa and 0.7 MPa respectively. The curves of purple, pink and blue show the results at different specific surface area. When the porous medium is used, the pressures are declined to 1.02 MPa, 0.91 MPa and 0.86 MPa respectively.

Heat transfer surface area has a great effect on the compression efficiency. Fig. 19 shows the compression efficiency at compression ratio of 7. The red curve shows the adiabatic compression efficiency is 66%. As heat transfer surface area increases, the compression efficiency increases. When the porous medium is used, the compression efficiency reached 85.4%, 90.9% and 93.5% respectively. The blue curve shows the negative influence on the efficiency by the resistance lies between 0.1 and 3%. As the heat transfer surface area S_{vex} increases, the resistance has greater influence on the efficiency. It is reasonable because the inner passage of the porous medium becomes small. Fig. 20 shows the work done by the resistance take up 4% compared with the compression work.

4.5. Speed n

From Eq. (22), the rotation speed is related to parameter Ka . Fig. 21 compares the effect of compression speed on the compression efficiency at compression ratio of 7. The red curve shows

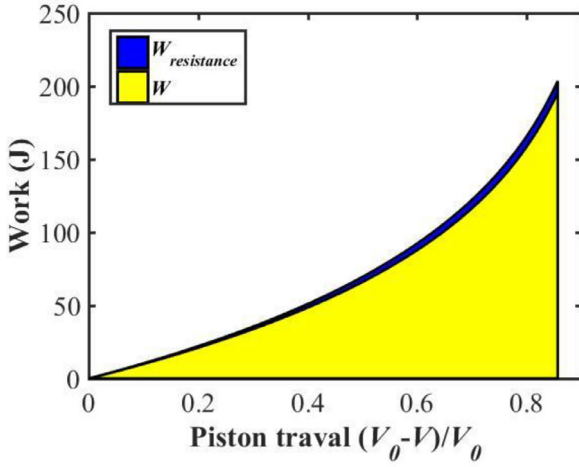


Fig. 20. Work done by resistance when $S_{\text{vex}} = 29.8$, $n = 120$ r/min.

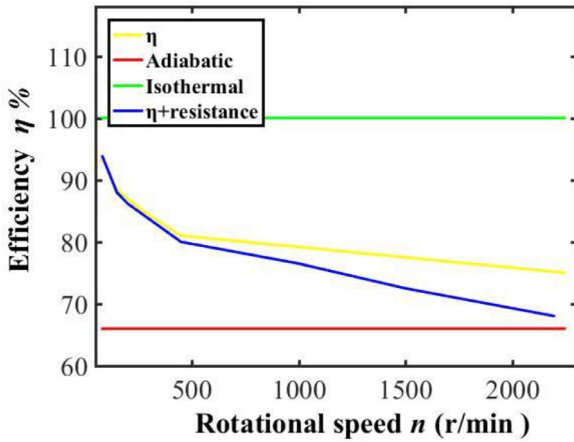


Fig. 21. Influence of parameter n on the compression efficiency.

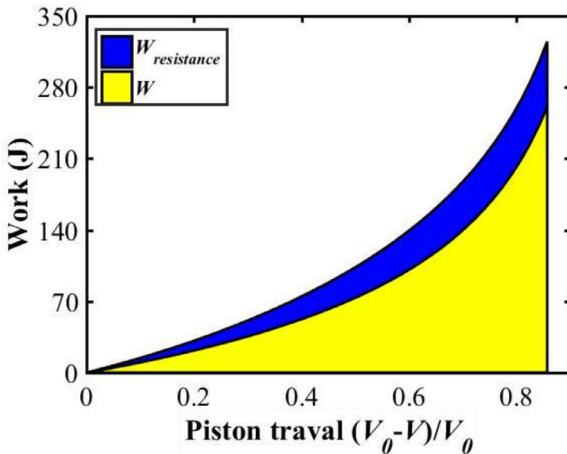


Fig. 22. Work done by resistance when $S_{\text{vex}} = 29.8$, $n = 1200$ r/min.

the adiabatic compression efficiency is 66%. When the compression speed declines from 2200 to 100 r/min, the efficiency increase from 74 to 96%. The blue curve shows the resistance has significant influence on efficiency from 0.1 to 11% with the increase in rotational speed from 100 to 2200. Fig. 22 shows the work done by the resistance take up 15% when the speed reaches to 1200 r/min.

5. Conclusions

Adiabatic compression allows the heat to be accumulated in the air. The work passed into a compressor is magnified by the heat generated from air compression when a same amount of air passed through a compressor. For adiabatic CAES systems, the heat is passed into storage through a heat exchange medium to improve the turnaround efficiency. For ICAES systems, the heat passed through the heat exchange medium is reduced because less of the work is passed into the compressor. The exergy loss in the heat exchange medium will be reduced as smaller amount of the heat is passed through it. In this study, an isothermal piston compression is proposed to improve the efficiency of compressors and expanders by achieving isothermal compression. This method is to manage the heat from air compression. A dimensionless analysis is performed to the equations of the system. The equations are simplified, as number of the parameters of the equations becomes less with the definition of the parameters Ka and Xu . These definitions provide a new approach to look at the thermodynamics of the compressor. Conclusions are made as following:

- 1 Ka measures the ratio between the heat transferred from the air and the heat generated in the air. Numbers of Ka quantify the extent of the compression approaching isothermal. $Ka = 0$ means non-isothermal compression. $Ka = \infty$ means isothermal compression. When increasing the number of Ka , the compression approaches isothermal. When Ka is over 80, the temperature of the air is reduced by 80% compared with this is for adiabatic compression.
- 2 Xu measures the heat passed through the porous medium per unit of the heat capacity. Numbers of Xu quantify the level of the temperature of the porous medium being changeable. $Xu = 0$ means that the temperature is not change. When increasing the number of Xu , the temperature of the porous medium will change with that of the air. For general porous mediums, the change of the temperature of the porous medium is not significant. The equation for the temperature of the air dominates the thermodynamic characteristics of the compressor. The thermodynamic information is mainly contained in the parameter Ka .

Ka and Xu bring the factors (heat transfer coefficient, properties of porous material, volume of compression chamber and rotational speed) that influence compression together so that these influences can be analyzed comprehensively. The parameters are useful for system optimization.

- 1 The reported results show the proposed isothermal piston is effective in improving the efficiency of the compressor. The growth of the compression efficiency increases with the compression ratio. As the compression ratio is over 7, the growth of compression efficiency gradually becomes flat. With the porous material (specific surface area 27.4 cm^{-1} , porosity 0.92), the compression efficiency increases by 11% ($n = 1200$) and 28% ($n = 120$) at the compression ratio of 7.
- 2 A portion of the heat generated from air compression will be accumulated in the porous medium, when the compressor works for a long time. The equilibrium temperature of the porous medium will be higher than the liquid temperature. This can lead to a decline in the heat transfer between the air and the porous medium. The equilibrium temperature can be reduced by increasing the surface area $S_{\text{por}0}^*$ of the porous medium that is immersed in the liquid.

First prototype of the compressor is being manufactured currently. There are still some practical problems associated. First analysis will be to prove that the porous medium can affect the temperature and the pressure of the air in compression chamber

as the time for heat transfer is very short. The porous material applied in the experiments is a foamed metal copper. Foam metals are commonly used engineering material. These materials both have nice conduction of metals and great surface area of porous mediums. The porous material is manufactured in disk shape. Several pieces of the disk-shaped porous material are stacked and are mounted on the piston rod under the piston. Experiments have been performed at speed of 60 r/min (time for compression is ~0.4 s) and at the compression ratio of 2–4. The case applying porous mediums is compared with the case without applying porous mediums. Pressures of the air approach isothermal compression when the porous medium is used.

Measurement of temperatures of the air is challenging because the temperature changes at a high frequency (~2.5 Hz). According to Eq. (1), the temperature of the air is related to the pressure and the volume. Applying Eq. (1) to the air in initial condition and in compressed condition, Eq. (25) is obtained showing that the temperature, the pressure, and the volume of the air in compressed condition is related to that is in the initial condition. Initial temperature (θ_{air0}), pressure (p_0), and volume (V_0) are measured before applying compression. Initial mass (m_0) of the air is calculated by ($p_0 \times V_0 / R / \theta_{air0}$). Pressure sensor and position encoder are used to measure the pressure (p) and the volume ($V = V_0 - \text{Area} \times \text{position}$) during air compression.

$$\theta_{air} = \frac{pV}{p_0V_0} \cdot \frac{m_0}{m} \cdot \theta_{air0} \quad (25)$$

Air pressure increases as compression proceeds, causing air to dissolve into water. Moreover, the mass of the air decreases with increases in the pressure. According to Henry's law, the amount of dissolved gas in a liquid is proportional to its partial pressure above gas:

$$c = H^{cp} \cdot p \quad (26)$$

where, c (kg/m^3) is the mass of gas dissolved in the liquid per unit of volume of the liquid; p is the gas pressure above the liquid; H^{cp} ($\text{kg}/\text{m}^3/\text{Pa}$) is Henry's Law constant.

The mass of the air in the compression chamber is obtained,

$$m = m_0 - V_{liq} \cdot c \quad (27)$$

The temperature of the air can be measured with Eqs. (25–27).

The next step will be to validate the theory at higher piston speed and higher pressure ratio. As pressure ratio increases, a greater volume of air will be dissolved in water. Both heat transfer and dissolution will affect the pressure and the temperature of the air. Henry's law can be applied at equilibrium state or at a low piston speed, but cannot be used at a high speed. Dealing with dissolution is challenging. As this is a theoretical study, details about the experiments are not mentioned in this paper.

Declaration of Competing Interest

None.

CRediT authorship contribution statement

Xu Weiqing: Conceptualization, Methodology. **Du Ziyue:** Software, Data curation, Writing - original draft. **Wang Xiaoshuang:** Visualization, Investigation. **Cai Maolin:** Supervision, Funding acquisition. **Jia Guanwei:** Writing - review & editing. **Shi Yan:** Project administration.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.jheatmasstransfer.2020.119779.

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